

Finding the Task-Optimal Low-Bit Sub-Distribution in Deep Neural Networks

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Abstract

Quantized neural networks typically require smaller memory footprints and lower computation complexity, which is crucial for efficient deployment. However, quantization inevitably leads to a distribution divergence from the original network, which generally degrades the performance. To tackle this issue, massive efforts have been made, but most existing approaches lack statistical considerations and depend on several manual configurations. In this paper, we present an adaptive-mapping quantization method to learn an optimal latent sub-distribution that is inherent within models and smoothly approximated with a concrete Gaussian Mixture (GM). In particular, the network weights are projected in compliance with the GM-approximated sub-distribution. This sub-distribution evolves along with the weight update in a co-tuning schema guided by the direct task-objective optimization. Sufficient experiments on image classification and object detection over various modern architectures demonstrate the effectiveness, generalization property, and transferability of the proposed method. Besides, an efficient deployment flow for the mobile CPU is developed, achieving up to $7.46\times$ inference acceleration on an octa-core ARM CPU. Our codes have been publicly released at <https://github.com/RunpeiDong/DGMS>.

1. Introduction

Deep neural networks (DNNs) have become the *de-facto* method in plentiful applications, including computer vision (He et al., 2016; 2020), natural language processing (Devlin et al., 2019; Xu et al., 2020), speech recognition (Hinton et al., 2012), and robotics (Kim & Moon, 2016; Liu et al., 2016a). Although DNNs are pushing the

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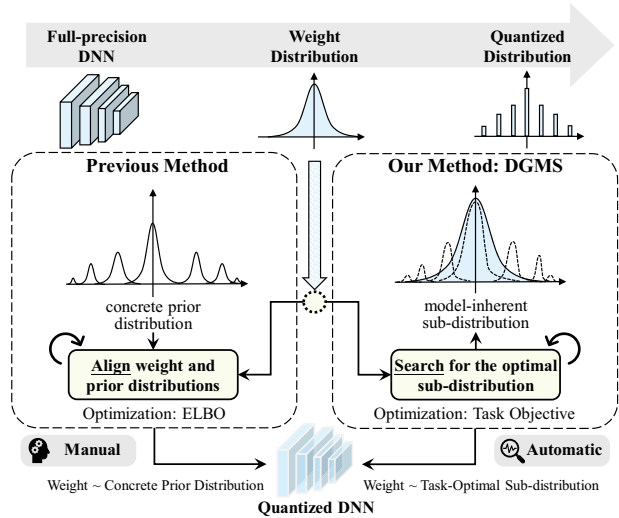


Figure 1: Comparison between the previous methods and our proposed DGMS. The training objective of the previous methods is ELBO while it is the task-objective only for DGMS, and the previous methods depend on manual configurations while our DGMS is fully automated.

limits of generalization performance across various areas, the growing memory and computation severely hinder the practical deployment. Abundant acceleration and compression methods have been proposed to tackle this issue, such as pruning (LeCun et al., 1989; Han et al., 2016), weight quantization (Fiesler et al., 1990; Rastegari et al., 2016), knowledge distillation (Hinton et al., 2015), efficient model design (Ma et al., 2018; Han et al., 2020; Tan et al., 2021), and even neural architecture search (Zoph et al., 2018).

Among the aforementioned schemes, weight quantization, as one of the mainstream techniques, aims to eliminate the representative redundancy via shortened bit-width. Generically, quantization can be represented as a projection $Q : \mathcal{W} \in \mathbb{R} \rightarrow \mathcal{Q} = \{q_0, q_1, \dots, q_K\}$, where $\mathcal{W} \sim \mathcal{D}_{\mathcal{W}}$ is the real-valued preimage while $\mathcal{Q} \sim \mathcal{D}_{\mathcal{Q}}$ denotes the compressed and discrete representation. The categorical distribution $\mathcal{D}_{\mathcal{Q}}$ can be viewed as a factorized Dirac distribution for the expected precision, which inherently suffers a divergence from the preimage $\mathcal{D}_{\mathcal{W}}$. Consequently, the weight deviation after low-bit quantization generally leads to significant performance degradation.